Posture Detection using IMU Sensor and Neural Network-Based Classification

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A. System Design

This project was developed to design a posture detection system capable of classifying static human postures using inertial measurement data collected from an Arduino Nano 33 BLE Sense Rev2. The board's integrated BMI270/BMM150 IMU sensor provides triaxial accelerometer and gyroscope readings that describe the board's linear acceleration and rotational orientation in real-time. These readings were analyzed using a neural network trained to identify five primary postures: Supine, Prone, Side, Standing, and Unknown.

The system architecture follows a modular design with two tightly coupled components: (1) a data acquisition module implemented on the Arduino microcontroller, responsible for capturing acceleration and orientation data, and (2) an offline machine learning module built in Python, used for preprocessing, training, and testing posture classification models. The Arduino continuously samples IMU data at approximately 50–100 Hz, achieving a balance between capturing rapid orientation changes and avoiding redundant samples. Each measurement is timestamped, formatted, and transmitted to a Python-based serial logger that records data only after receiving an 'r' command from the user. This controlled logging prevents unnecessary data accumulation and organizes the dataset by posture category.

Every row in the dataset includes: time_ms, ax, ay, az, pitch, roll, dotS, dotL, and label. These variables together provide a complete snapshot of the sensor's orientation and motion state. The inclusion of derived quantities such as dotS and dotL—dot products relative to reference axes—enables the model to capture directional cosine relationships, which proved effective in distinguishing similar postures like prone and supine.

During early experiments, it became clear that the model's accuracy was highly dependent on the physical orientation of the IMU sensor. Even minor changes in board alignment produced large variations in readings. To mitigate this, data were collected twice for each posture—once with the board's USB port facing upward and once downward. This approach introduced orientation invariance during training, allowing the neural network to recognize postures irrespective of board alignment.

Gaussian noise (σ=0.02) was also added to simulate real-world micro-motions caused by vibrations or user breathing patterns.

The motivation behind this project extends to several healthcare-related applications: posture monitoring in elderly patients, fall detection systems, and rehabilitation feedback systems. By leveraging a small, low-power IMU board and efficient neural modeling, this project demonstrates the feasibility of achieving highly accurate posture recognition without complex hardware or costly sensors.

B. Experiment

The experimental phase involved careful collection of IMU data under controlled laboratory conditions. Each posture was simulated for 2–3 minutes on a flat surface, generating approximately 4,000–6,000 samples per posture. The experimental setup maintained consistent ambient lighting, temperature, and surface stability to ensure that only gravitational and orientation changes influenced the sensor readings.

Five postures were defined for classification: Supine (lying face up), Prone (lying face down), Side (lying on either side), Standing (vertical position), and Unknown (irregular movements or transitions). The Unknown class was crucial for realistic modeling, representing cases when the sensor experiences motion not belonging to any specific posture. These instances included shaking, tilting, or repositioning the board.

Each dataset was standardized and merged to create a unified CSV file. Z-score normalization was applied to remove bias caused by feature magnitude differences. This ensured that the neural network could learn feature relationships purely from orientation behavior rather than numerical scale. Outliers or corrupted readings were removed through NaN filtering and threshold-based clipping of physically implausible accelerations ($|a| > 2g$).

Noise augmentation was applied during preprocessing to mimic the sensor instability typically observed in wearable applications. This method significantly improved model generalization, ensuring that the classifier could maintain accuracy even in less controlled settings. By diversifying the data with slight perturbations, the model effectively learned more robust posture boundaries in feature space.

C. Algorithm

The posture classification algorithm employed a feedforward neural network built using TensorFlow and Keras. The model architecture was chosen for simplicity and computational efficiency, as the feature space contained only seven input parameters. The neural network consisted of two hidden layers with 128 and 64 neurons respectively, each using one of three tested activation functions: Sigmoid, Tanh, and ReLU. Dropout regularization (rate = 0.3) and Batch Normalization were integrated after each dense layer to stabilize gradient flow and prevent overfitting.

The training process utilized the Adam optimizer with an initial learning rate of 0.001 and categorical cross-entropy loss. The model was trained in mini-batches of 128 samples, and early stopping with patience of 10 epochs was applied to halt training when validation accuracy ceased improving. A validation split of 20% ensured that performance was evaluated on unseen data during each epoch.

Feature scaling and normalization were critical for stability. Because IMU readings often vary in units and magnitudes across axes, unscaled data would bias the optimization process. The standardized features enabled uniform gradient magnitudes, leading to faster and more consistent convergence. Additionally, Gaussian noise in the feature space served as a regularizer that improved the model's ability to handle sensor drift.

Empirically, activation function comparison revealed that ReLU provided the fastest convergence and highest final accuracy due to its non-saturating gradient property. Sigmoid and Tanh, while producing smooth transitions, occasionally suffered from vanishing gradients when the feature distribution widened. Nevertheless, all three functions achieved extremely high performance, with validation accuracies surpassing 99.9%. This demonstrates that the IMU-based posture problem is linearly separable to a significant degree when sufficient feature preprocessing is applied.

D. Results

The neural network achieved outstanding results across all activation configurations. Validation accuracies reached above 99.9% within the first 10 epochs, and losses converged to near zero. Specifically, Sigmoid achieved 99.97%, Tanh achieved 99.98%, and ReLU achieved 99.99% validation accuracy. These near-perfect results indicate that the features extracted from IMU data capture distinctive patterns corresponding to each posture.

Training and validation loss curves confirmed stable convergence without oscillations, demonstrating effective regularization and absence of overfitting. Dropout and early stopping mechanisms ensured the model did not memorize the dataset but learned underlying posture-specific relationships. All postures—Supine, Prone, Side, Standing, and Unknown—were classified correctly, yielding a confusion matrix with a perfect diagonal structure. Both precision and recall for all classes were 100%, confirming that the model neither produced false positives nor false negatives in the test phase.

A deeper examination of the results reveals that the system benefited from orientation normalization and balanced data representation. Without the mirrored data collection (USB-up/down), initial experiments showed bias toward certain orientations, reducing classification accuracy for prone and side postures. The inclusion of mirrored samples eliminated this bias completely. The synthetic noise also contributed by preventing the model from overfitting to the exact sensor environment of training, leading to generalizable behavior during testing.

The high consistency across all activation types indicates that the network's decision boundaries are strongly defined and largely unaffected by activation nonlinearity differences. However, ReLU's faster convergence and minimal computational overhead make it the ideal choice for potential microcontroller deployment in future iterations of the system.

E. Discussion

The experimental and algorithmic results validate the hypothesis that a neural network can accurately distinguish between static postures using IMU-derived features. The combination of accelerometer, gyroscope, and orientation-derived quantities enables precise estimation of the board's spatial attitude. The project highlights how cost-effective embedded systems can provide high-fidelity activity recognition without requiring external cameras or additional sensors.

The main challenges encountered were orientation bias, sensor drift, and limited real-world variability. Orientation bias was mitigated by collecting data in multiple physical orientations, while sensor drift was handled through normalization and noise regularization. Limited variability was addressed through data augmentation, although future iterations could involve real human subjects or wearable configurations to introduce natural variability in body dynamics.

Potential extensions of this project include deployment using TensorFlow Lite for Microcontrollers, which would enable real-time posture inference directly on the Arduino Nano 33 BLE Sense. Additional improvement opportunities lie in detecting transitional states—such as sitting-to-standing or rolling transitions—by integrating time-series processing with models like LSTM or 1D CNN. Moreover, introducing adaptive thresholding for the Unknown class could make the system more resilient in unpredictable environments.

In conclusion, this project successfully demonstrates an efficient end-to-end posture detection system leveraging embedded sensing and neural network modeling. The combination of clean data acquisition, robust preprocessing, and optimized training resulted in a system capable of near-perfect classification accuracy. Such a framework can serve as a foundation for more advanced wearable healthcare monitoring systems or smart rehabilitation platforms.

References

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